

Verification Test Cases for the Linear Buckling Analysis in 4C

Three-dimensional solid elements

Linear buckling feature, branch `dev-linear-buckling` of `m-frey/4C`

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Overview

This document defines four test cases for verifying the linear buckling analysis of 3D solid elements in 4C. For each case it gives the setup, the derivation of the reference solution, and the accepted deviation. All numbers to be matched are collected in Chapter 5.

Case	Structure	Reference	Tests mainly
A	Simply supported plate, uniaxial compression	analytical (Kirchhoff, Mindlin)	plate modes, several (m, n) , Neumann load
B	Cantilever column, rectangular section	analytical (Euler, Timoshenko)	mode ordering, preload path
C	3D beam-like solid, clamped–pinned	published benchmark (FEniCSx)	code-to-code, identical mesh
D	Periodic square-lattice cell	plausibility + exact identities	periodic BCs, Dirichlet load

Eigenvalue problem and notation. All cases solve

$$(\mathbf{K}_T(\mathbf{u}_0) + \lambda \mathbf{K}_\Delta) \phi = \mathbf{0}, \quad (1)$$

where $\mathbf{u}_0$ is the base state ($\mathbf{u}_0 = \mathbf{0}$ unless stated otherwise), $\mathbf{K}_\Delta$ is the differential (initial-stress) stiffness due to a reference load pattern P_{ref} , and the critical load is $P_{\text{cr}} = P_{\text{pre}} + \lambda P_{\text{ref}}$. Positive λ means the load acts as given; negative λ means it acts in reverse.

Differential stiffness. 4C uses the classical geometric stiffness $\mathbf{K}_\Delta = \mathbf{K}_\sigma(\Delta \mathbf{S})$ of the perturbation stress only, as Abaqus *BUCKLE and the FEniCSx benchmark (Case C) do (design choice D1 of the implementation plan). A finite-difference tangent, $\mathbf{K}_\Delta \approx [\mathbf{K}_T(\mathbf{u}_0 + \varepsilon \Delta \mathbf{u}) - \mathbf{K}_T(\mathbf{u}_0)]/\varepsilon$, would additionally contain the initial-displacement term (linear in $\nabla \Delta \mathbf{u}$). Its relative effect on λ_i is of the order of the pre-buckling strain at the critical point, $\varepsilon_{\text{pre},i} \approx \sigma_{\text{cr},i}/E$. This is why that variant was rejected. The tolerances below apply to the classical $\mathbf{K}_\Delta$.

Mode identification and comparison.

- Modes are identified by their dominant displacement component and by counting half-waves along the edges or axes.
- The shape is compared with the analytical mode ϕ^{an} at the nodes of the mid-surface or neutral axis, using the modal assurance criterion $\text{MAC} = (\phi^{\text{T}} \phi^{\text{an}})^2 / ((\phi^{\text{T}} \phi)(\phi^{\text{an T}} \phi^{\text{an}}))$.
- For degenerate eigenvalues (multiplicity > 1), only the eigenvalue is compared, together with the fact that the mode lies in the analytical eigenspace.

Units. Cases A, B and D use mm, N and MPa. Case C uses the dimensionless units of the published benchmark.

1 Case A – Simply supported plate under uniaxial compression

1.1 Problem definition

A flat rectangular plate of length a (in x), width b (in y) and thickness t (in z) is simply supported on all four edges and loaded by a uniform compressive membrane force N_x per unit width on the edges $x = 0$ and $x = a$. The plate is modelled as a 3D solid; no periodic boundary conditions are used.

Quantity	A1 (square)	A2 (long)
Length a	100 mm	300 mm
Width b	100 mm	100 mm
Thickness t	1 mm	1 mm
Young's modulus E	210 000 MPa	
Poisson's ratio ν	0.3	
Material	St. Venant–Kirchhoff, KINEM nonlinear	
Reference load N_{ref}	1 N/mm, i.e. traction $N_{\text{ref}}/t = 1$ MPa	

Configuration A1 has well-separated eigenvalues and is the primary test. Configuration A2 has several closely spaced eigenvalues with different half-wave numbers m . It tests that the eigensolver finds the correct eigenvalues and returns them in the correct order.

1.2 3D model and boundary conditions

A simple support in the 3D solid has to act on the mid-surface $z = t/2$ only. Fixing w on the whole edge face would also restrain the rotation, i.e. partially clamp the edge.

1. **Out-of-plane support:** $u_z = 0$ on the four mid-surface edge lines ($z = t/2$ at $x = 0$, $x = a$, $y = 0$, $y = b$). This requires a node layer at $z = t/2$, i.e. an even number of element layers, or quadratic elements with mid-thickness nodes.
2. **Loading:** uniform dead surface traction $p = N_{\text{ref}}/t$ on both loaded faces $x = 0$ and $x = a$, acting inward (compression). The load is self-equilibrated.
3. **Remaining in-plane rigid-body modes (statically determinate):**
 - $u_x = 0$ and $u_y = 0$ at point $P_1 = (0, 0, t/2)$;
 - $u_y = 0$ at point $P_2 = (a, 0, t/2)$.

The pre-buckling state is then exactly $\sigma_{xx} = -N_x/t$, $\sigma_{yy} = \sigma_{xy} = 0$. The unloaded edges are free in their plane and Poisson expansion is unrestrained.

4. **Elements:** SOLID HEX20 or HEX27 with at least 2 layers through the thickness, or HEX8 with EAS (`eas_full`) and at least 2 layers. Plain HEX8 locks severely at $b/t = 100$ and must not be used.
5. **Recommended in-plane mesh:** at least 8 quadratic elements per half-wave. That is 20×20 for A1 and 60×20 for A2 (the highest listed mode of A2 has $m = 8$).

1.3 Analytical solution (Kirchhoff plate theory)

The bending stiffness is $D = Et^3/[12(1 - \nu^2)]$. Linearized about the membrane state N_x (compression positive), the governing equation is

$$D \nabla^4 w + N_x \frac{\partial^2 w}{\partial x^2} = 0. \quad (1.1)$$

The simply supported boundary conditions ($w = 0$ and $M_n = 0$ on all edges) are satisfied exactly by the Navier ansatz $w_{mn} = \sin(m\pi x/a) \sin(n\pi y/b)$ with $m, n \in \mathbb{N}$. Substituting it gives

$$N_{x,mn} = \frac{\pi^2 D}{b^2} k_{mn}, \quad k_{mn} = \left(\frac{mb}{a} + \frac{n^2 a}{mb} \right)^2. \quad (1.2)$$

The eigenvalues are $\lambda_{mn} = N_{x,mn}/N_{\text{ref}}$, and w_{mn} is the corresponding buckling mode, i.e. the out-of-plane displacement of the mid-surface. For $a/b = 1$ the minimum is $k_{11} = 4$. For $a/b = 3$ it is $k_{31} = 4$, with the neighbours $k_{41} = 4.340$, $k_{21} = 4.694$, $k_{51} = 5.138$.

Transverse-shear correction (Mindlin plate theory). A 3D solid includes transverse-shear flexibility. For a simply supported Mindlin plate, the exact relation to the Kirchhoff value is

$$N_{x,mn}^{\text{M}} = \frac{N_{x,mn}}{1 + D \kappa_{mn}^2 / (\kappa G t)}, \quad \kappa_{mn}^2 = \left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2, \quad G = \frac{E}{2(1 + \nu)}, \quad \kappa = \frac{5}{6}. \quad (1.3)$$

For $b/t = 100$ the correction is below 0.6 % for all listed modes. The Mindlin value is the primary reference for the 3D model.

1.4 Expected deviations of the 3D model

- Remaining 3D effects (thickness change, edge zones of the line support) are of order $(t/b)^2$ and negligible.
- The discretization error of conforming displacement elements usually overestimates λ . It decreases under mesh refinement and dominates the total deviation.
- Pre-buckling strain: $\varepsilon_{\text{pre}} \approx 75.9/210000 \approx 3.6 \times 10^{-4}$ (mode 1), so geometric nonlinearity of the pre-buckling state is negligible.

2 Case B – Cantilever column with rectangular cross-section

2.1 Problem definition

A straight prismatic bar of length L with rectangular cross-section $b \times h$ (b in y , h in z , $h < b$) is clamped at $x = 0$ and loaded at $x = L$ by a dead axial compressive force P .

Length L	100 mm
Cross-section $b \times h$	3 mm $\times$ 2 mm, $A = 6 \text{ mm}^2$
E, ν	210 000 MPa, 0.3
Material	St. Venant–Kirchhoff, KINEM nonlinear
Reference load P_{ref}	1 N, i.e. traction 1/6 MPa on the end face

2.2 3D model and boundary conditions

- **Clamped end:** $u_x = u_y = u_z = 0$ on all nodes of the face $x = 0$. This also blocks Poisson contraction there, a local effect of order h/L on the stress field that is negligible for λ at $L/h = 50$.
- **Free end:** uniform dead traction $-P_{\text{ref}}/A$ in x direction on the face $x = L$. There are no further boundary conditions.
- **Elements and mesh:** SOLID HEX20 or HEX27 with at least 2×3 elements in the cross-section and at least 50 along L . Alternatively HEX8 with EAS, $4 \times 6 \times 100$.

2.3 Analytical solution

Euler–Bernoulli. From $EI v'''' + P v'' = 0$ with $v(0) = v'(0) = 0$ and $M(L) = Q(L) = 0$:

$$P_{E,n} = \frac{(2n-1)^2 \pi^2 EI}{4L^2}, \quad v_n(x) = 1 - \cos \frac{(2n-1)\pi x}{2L}, \quad n = 1, 2, \dots \quad (2.1)$$

This holds separately for both bending axes:

- weak axis, deflection in z : $I_y = bh^3/12 = 2 \text{ mm}^4$;
- strong axis, deflection in y : $I_z = hb^3/12 = 4.5 \text{ mm}^4$.

The ratio $I_z/I_y = 2.25$ interleaves the two families, so the mode order tests that modes are ordered correctly. Torsional buckling of a solid rectangular section is far above the listed range.

Shear correction (Engesser).

$$P_{T,n} = \frac{P_{E,n}}{1 + P_{E,n}/(\kappa GA)}, \quad \kappa = \frac{10(1+\nu)}{12+11\nu} = 0.8497 \text{ (Cowper)}. \quad (2.2)$$

The Haringx variant differs from Engesser by less than 0.02 % for all listed modes. $P_{T,n}$ is the primary reference.

2.4 Additional check: preloaded base state

1. Run the nonlinear static preload with $P_{\text{pre}} = 0.5 P_{T,1}$ (e.g. NUMSTEP: 5).
2. Then run the buckling step with the same P_{ref} .

The pre-buckling path of this structure is practically linear ($\varepsilon_{\text{pre}} < 10^{-4}$), so the result must satisfy $P_{\text{pre}} + \lambda_1^{(\text{pre})} P_{\text{ref}} = \lambda_1^{(0)} P_{\text{ref}}$ up to $\approx 10^{-3}$ relative.

3 Case C – Published benchmark: FEniCSx 3D solid

3.1 Source

J. Bleyer, M. Pierre, *Linear Buckling Analysis of a 3D solid*, Numerical Tours of Computational Mechanics with FEniCSx, https://bleyerj.github.io/comet-fenicsx/tours/eigenvalue_problems/buckling_3d_solid/buckling_3d_solid.html. The source is in <https://github.com/bleyerj/comet-fenicsx>, directory `tours/eigenvalue_problems/buckling_3d_solid`, commit 5d06010c (4 March 2025).

The tour solves exactly the classical problem $\mathbf{K}\phi = -\lambda\mathbf{K}_\sigma\phi$ with 3D solid elements, where $\mathbf{K}$ is the linear stiffness and $\mathbf{K}_\sigma$ is built from the linear pre-stress σ_0 .

3.2 Problem definition (identical to the benchmark)

Domain	box $[0, L] \times [0, b] \times [0, h]$, $L = 1$, $b = 0.01$ (in y), $h = 0.03$ (in z)
Material	isotropic linear elastic, $E = 10^3$, $\nu = 0.3$
Mesh	$50 \times 5 \times 5$ hexahedra, 2nd-order Lagrange (Q_2 , 27 nodes) $\Rightarrow$ SOLID HEX27
DOFs	36 663 (including constrained DOFs)
Left end $x = 0$	fully clamped: $u_x = u_y = u_z = 0$ on all face nodes
Right end $x = L$	$u_y = u_z = 0$ on all face nodes, u_x free
Load	dead traction $\mathbf{t} = (-N_0, 0, 0)$ on the face $x = L$, $N_0 = 1$ (a stress)
Output	eigenvalue $\lambda =$ critical <i>axial stress</i> ; critical force $F_{cr} = \lambda b h$

Replicating it in 4C.

- Generate the same structured $50 \times 5 \times 5$ HEX27 mesh.
- Boundary conditions:
 - DESIGN SURF DIRICH on $x = 0$ with ONOFF [1,1,1];
 - DESIGN SURF DIRICH on $x = L$ with ONOFF [0,1,1];
 - DESIGN SURF NEUMANN on $x = L$ with VAL [-1,0,0].
- Material MAT_Struct_StVenantKirchhoff with $E = 1000$, $\nu = 0.3$, KINEM nonlinear, base state $\mathbf{u}_0 = \mathbf{0}$.

At $\mathbf{u}_0 = \mathbf{0}$ the 4C tangent $\mathbf{K}_T(\mathbf{0})$ equals the benchmark's $\mathbf{K}$.

3.3 Analytical solution (beam theory, as in the tour)

Clamped–pinned Euler column: $EI v'''' + Fv'' = 0$ with $v(0) = v'(0) = 0$ and $v(L) = v''(L) = 0$. This leads to the transcendental equation

$$\tan \alpha_n = \alpha_n, \quad F_{cr,n} = \alpha_n^2 \frac{EI}{L^2}, \quad \lambda_n = \frac{F_{cr,n}}{b h}. \quad (3.1)$$

The exact roots are $\alpha_n/\pi = 1.430297, 2.459024, 3.470890, 4.477409, 5.481537, 6.484387$. The two bending axes have

- weak axis, deflection in y : $I = hb^3/12$;
- strong axis, deflection in z : $I = bh^3/12 = 9 \times$ weak.

The shear-corrected (Engesser) values are listed as well.

Two remarks on the published tour.

1. The tour quotes $\alpha_2 \approx 2.4509\pi$. The correct root of $\tan \alpha = \alpha$ is $\alpha_2 = 2.459024\pi$.

2. The tour compares FE mode i with the i -th weak-axis beam mode for all i . FE modes 4 and 8, however, are strong-axis modes (dominant u_z), so from mode 4 on the tour's comparison pairs the wrong modes. The table in Chapter 5 assigns each mode to its beam counterpart.

3.4 Numerical reference values

The FE reference values were produced by running the unmodified benchmark script. The only changes were an API update for dolfinx 0.9 (`.vector` $\rightarrow$ `.x.petsc_vec`) and additional output. The run used the official container `dolfinx/dolfinx:v0.9.0` with SLEPc Krylov–Schur, shift-invert $\sigma = 0.01$, tolerance 10^{-12} .

- The published mesh $50 \times 5 \times 5$ (36 663 DOFs) is the primary reference for the code-to-code comparison.
- The refined mesh $80 \times 8 \times 8$ (139 587 DOFs) is the converged reference.
- A $100 \times 5 \times 5$ run (72 963 DOFs) differs from $80 \times 8 \times 8$ by at most 4×10^{-4} relative for modes 1–8, which indicates the remaining discretization error.
- A $100 \times 10 \times 10$ run exceeded the 16 GB of memory available and was not used.

The FE eigenvalues are not monotone under refinement, and mode 1 slightly increases. This is expected, because the discrete pre-stress enters $\mathbf{K}_\sigma$, so conforming FE eigenvalues are not bounds here. All eight modes have at least 99.8% of their norm in the dominant component (u_y weak axis, u_z strong axis).

4 Case D – Periodic square-lattice cell (plausibility)

4.1 Purpose

There is no clean analytical solution for a buckling problem that 4C's periodic RVE conditions can represent: they require periodicity on all six faces of a cuboid cell. This case is therefore checked by

1. exact identities that any correct implementation must satisfy, and
2. a closed-form frame-theory estimate as a plausibility band.

It is the only case that exercises periodic boundary conditions and a Dirichlet (prescribed macro-strain) perturbation load.

4.2 Problem definition

- **Unit cell:** $[0, L] \times [0, L] \times [0, d]$ with $L = 20$ mm, $d = 1$ mm, and a square hole $[w/2, L - w/2]^2$ through the full depth, $w = 1$ mm. The material thus forms a square frame whose struts of total thickness w are split in two halves across the periodic faces. Tiling gives an infinite square lattice with strut spacing L and strut thickness w .
- **Material:** St. Venant–Kirchhoff, $E = 210\,000$ MPa, $\nu = 0.3$.
- **Periodic BCs:** `DESIGN SURF PERIODIC RVE 3D BOUNDARY CONDITIONS` on all six faces, penalty enforcement (`CONSTRAINT_ENFORCEMENT: penalty`), automatic corner detection.
- **Macro loading via the corner reference nodes** (N1 at the origin; N2, N4, N5 the corners in $+x$, $+y$, $+z$):
 - N1: all components fixed;
 - N2: $u_x = -\varepsilon_{\text{ref}}L$ with $\varepsilon_{\text{ref}} = 10^{-3}$ through a time function, $u_y = u_z = 0$;
 - N4: $u_x = u_z = 0$, u_y free;
 - N5: $u_x = u_y = 0$, u_z free.

This gives uniaxial macro compression with free lateral contraction and no macro shear or rotation.

- **Two model sizes:**
 - D1: the $1 \times 1 \times 1$ cell;
 - D2: a $2 \times 2 \times 1$ super-cell with the identical mesh in every sub-cell.
- **Mesh:** structured, HEX20/HEX27, at least 2 elements across each half-strut, at least 40 along a strut, 1 element in z .
- **Serial run:** 4C's RVE conditions are not implemented in parallel.

4.3 Frame-theory estimate

The struts in x carry the whole load with strain ε . The struts in y are unloaded. Periodicity in z with free uniform ε_{zz} makes bending effectively plane strain, i.e. the bending stiffness is $Ew^3/[12(1 - \nu^2)]$ per unit depth. The joints are treated as rigid blocks of size w , so the free strut length is $\ell = L - w$.

- **D1 (period 1):** all joints rotate equally. The lowest mode is strut buckling between non-rotating joints (clamped–clamped, no sway):

$$\varepsilon_{\text{cr}}^{D1} = \frac{4\pi^2 w^2}{12(1 - \nu^2) \ell^2}. \quad (4.1)$$

- **D2 (period 2):** alternating transverse sway of neighbouring joint columns becomes admissible (clamped–guided):

$$\varepsilon_{\text{cr}}^{D2} = \frac{\pi^2 w^2}{12(1 - \nu^2) \ell^2} = \frac{1}{4} \varepsilon_{\text{cr}}^{D1}. \quad (4.2)$$

A periodic Euler–Bernoulli frame FE model (rigid joint zones, 16 elements per strut) reproduces both expressions to four significant digits. Its spectrum also contains every D1 eigenvalue within the D2 spectrum, which demonstrates identity (I1) below. Joint flexibility, shear and 3D effects are not included, so this estimate is only a plausibility band.

4.4 Exact identities (independent of any reference value)

- (I1) **Super-cell containment:** every eigenvalue of D1 appears in the spectrum of D2, possibly with higher multiplicity. The reason is that every 1-periodic field is also 2-periodic. Request at least $4\times$ as many modes for D2.
- (I2) **Monotonicity:** $\lambda_1^{D2} \leq \lambda_1^{D1}$.
- (I3) **Symmetry:** loading in y instead of x (swap the roles of N2 and N4) gives the same spectrum.
- (I4) **Periodicity of modes:** $\phi(\mathbf{x}^+) - \phi(\mathbf{x}^-)$ on paired nodes is of the order of the penalty error. It decreases by about $10\times$ when the penalty parameter is increased $10\times$.
- (I5) **Linearity in the load:** $\lambda(2\varepsilon_{\text{ref}}) = \lambda(\varepsilon_{\text{ref}})/2$ and $\lambda(-\varepsilon_{\text{ref}})$ flips the sign of the whole spectrum.
- (I6) **Penalty convergence:** $\lambda_1(\kappa)$ converges as $\kappa \rightarrow \infty$. The change from κ to 10κ must be at least $5\times$ smaller than the change from $\kappa/10$ to κ .

5 Reference results to match

This chapter collects all numbers and acceptance criteria. Relative deviation: $\delta_i = |\lambda_i^{4C} - \lambda_i^{\text{ref}}|/\lambda_i^{\text{ref}}$. The tolerances apply to the recommended meshes and are to be tightened into regression values once the 4C results have been verified.

5.1 Case A – simply supported plate

A1: $a = b = 100 \text{ mm}$, $t = 1 \text{ mm}$, $D = 19\,230.769 \text{ Nmm}$, $\pi^2 D/b^2 = 18.980\,008 \text{ N/mm}$

Mode	(m, n)	k_{mn}	Kirchhoff λ	Mindlin λ (reference)	Mindlin/Kirchhoff
1	(1,1)	4.000000	75.920034	75.877241	−0.056%
2	(2,1)	6.250000	118.625053	118.458034	−0.141%
3	(3,1)	11.111111	210.888983	210.295972	−0.281%
4	(2,2)	16.000000	303.680135	302.996602	−0.225%
5	(4,1)	18.062500	342.826403	341.190800	−0.477%
6	(3,2)	18.777778	356.402381	355.100634	−0.365%

λ = critical membrane force N_x in N/mm ($N_{\text{ref}} = 1 \text{ N/mm}$), equivalently the critical stress in MPa for $t = 1 \text{ mm}$. Mode 7 is degenerate: (1, 2) and (4, 2) both have $k = 25$. Do not compare modes beyond 6.

A2: $a = 300 \text{ mm}$, $b = 100 \text{ mm}$, $t = 1 \text{ mm}$

Mode	(m, n)	k_{mn}	Kirchhoff λ	Mindlin λ (reference)	Mindlin/Kirchhoff
1	(3,1)	4.000000	75.920034	75.877241	−0.056%
2	(4,1)	4.340278	82.378509	82.314032	−0.078%
3	(2,1)	4.694444	89.100595	89.064318	−0.041%
4	(5,1)	5.137778	97.515066	97.411294	−0.106%
5	(6,1)	6.250000	118.625053	118.458034	−0.141%
6	(7,1)	7.628118	144.781743	144.519114	−0.181%
7	(8,1)	9.251736	175.598030	175.197311	−0.228%
8	(1,1)	11.111111	210.888983	210.822928	−0.031%

Acceptance criteria

- A1 modes 1–3: $\delta_i \leq 1\%$; A1 modes 4–6: $\delta_i \leq 2\%$.
- A2 modes 1–8: $\delta_i \leq 1.5\%$, and the order of (m, n) must be exactly as listed. The spacing between modes 1 and 2 is only 8.5%, so a wrong order indicates an error.
- Mode shape: half-wave count (m, n) as listed; $\text{MAC} \geq 0.98$ against $\sin(m\pi x/a)\sin(n\pi y/b)$, evaluated with u_z at the mid-surface nodes.
- Mesh convergence: λ_1 decreases monotonically towards the Mindlin value under uniform refinement.
- Load reversal: reversing the traction gives $\lambda_i \rightarrow -\lambda_i$ (relative 10^{-8}). Doubling the traction halves λ_i .

5.2 Case B – cantilever column

$L = 100 \text{ mm}$, $b \times h = 3 \text{ mm} \times 2 \text{ mm}$; λ = critical force in N ($P_{\text{ref}} = 1 \text{ N}$).

Mode	Axis (deflection)	n	Euler P_E [N]	Timoshenko P_T [N] (reference)	P_T/P_E
1	weak (z)	1	103.630846	103.604771	−0.025%
2	strong (y)	1	233.169404	233.037442	−0.057%
3	weak (z)	2	932.677616	930.569806	−0.226%
4	strong (y)	2	2098.524636	2087.883909	−0.507%
5	weak (z)	3	2590.771155	2574.572274	−0.625%
6	weak (z)	4	5077.911464	5016.053142	−1.218%

Acceptance criteria

- Modes 1–3: $\delta_i \leq 1\%$; modes 4–6: $\delta_i \leq 2\%$.
- The order weak/strong/weak/strong/weak/weak must be exactly as listed. Dominant displacement component: u_z for weak-axis modes, u_y for strong-axis modes.
- Mode shape: $\text{MAC} \geq 0.98$ against $1 - \cos[(2n - 1)\pi x/(2L)]$, evaluated along the centroidal axis.
- Preload: with $P_{\text{pre}} = 51.80 \text{ N}$ ($\approx 0.5P_{T,1}$), $P_{\text{pre}} + \lambda_1^{(\text{pre})} \cdot 1 \text{ N}$ matches $\lambda_1^{(0)}$ within 10^{-3} relative.

5.3 Case C – FEniCSx benchmark

λ = critical compressive traction (stress) on the end face; $N_0 = 1$. The FEniCSx columns are the numerical reference; the beam columns are analytical plausibility values.

Mode	Axis	n	FEniCSx $50 \times 5 \times 5$	FEniCSx $80 \times 8 \times 8$	Euler beam	Timoshenko beam
1	weak (y)	1	0.167963169	0.168078306	0.16825607	0.16816949
2	weak (y)	2	0.496955439	0.496866498	0.49732930	0.49657360
3	weak (y)	3	0.987888956	0.987687419	0.99083224	0.98783717
4	strong (z)	1	1.500093336	1.500692451	1.51430464	1.50732006
5	weak (y)	4	1.642490988	1.640922961	1.64881509	1.64053797
6	weak (y)	5	2.455332621	2.451987909	2.47128677	2.45273882
7	weak (y)	6	3.429242204	3.421467271	3.45824987	3.42203704
8	strong (z)	2	4.392907156	4.392598990	4.47596370	4.41548720

Acceptance criteria

- **Code-to-code (primary):** 4C with HEX27, $50 \times 5 \times 5$, identical BCs, against FEniCSx $50 \times 5 \times 5$: $\delta_i \leq 10^{-4}$ for modes 1–8. The only allowed difference is the quadrature rule.
- Mode identification (dominant component) exactly as listed. The dominant component carries at least 99% of the mode norm.
- 4C $80 \times 8 \times 8$ HEX27 against FEniCSx $80 \times 8 \times 8$: same tolerances as above. Any finer 4C mesh against FEniCSx $80 \times 8 \times 8$: $\delta_i \leq 10^{-3}$.

5.4 Case D – periodic lattice cell

$L = 20 \text{ mm}$, $w = 1 \text{ mm}$, $\ell = L - w = 19 \text{ mm}$, $\varepsilon_{\text{ref}} = 10^{-3}$; $\lambda = \varepsilon_{\text{cr}}/\varepsilon_{\text{ref}}$.

Model	Mode type	Frame estimate λ	
		point joints, $\ell = L$	rigid joints, $\ell = L - w$
D1 ($1 \times 1 \times 1$)	clamped–clamped, no sway	9.0381	10.0145
D2 ($2 \times 2 \times 1$)	alternating sway	2.2595	2.5036

Acceptance criteria

- **Identities (hard):**
 - (I1) every D1 eigenvalue, up to the requested count, is found in the D2 spectrum within 10^{-6} relative (eigensolver tolerance);
 - (I2) $\lambda_1^{D2} < \lambda_1^{D1}$;
 - (I3) x - and y -loading spectra agree within 10^{-6} ;
 - (I4)–(I6) as defined in Chapter 4.
- **Plausibility (soft):**
 - λ_1^{D1} and λ_1^{D2} within $\pm 15\%$ of the rigid-joint frame estimate;
 - $\lambda_1^{D2}/\lambda_1^{D1}$ between 0.2 and 0.3 (frame theory: 0.25);
 - D1 mode: x -struts buckle with the cell's joints non-rotating; D2 mode: neighbouring joint columns sway in opposite y directions.